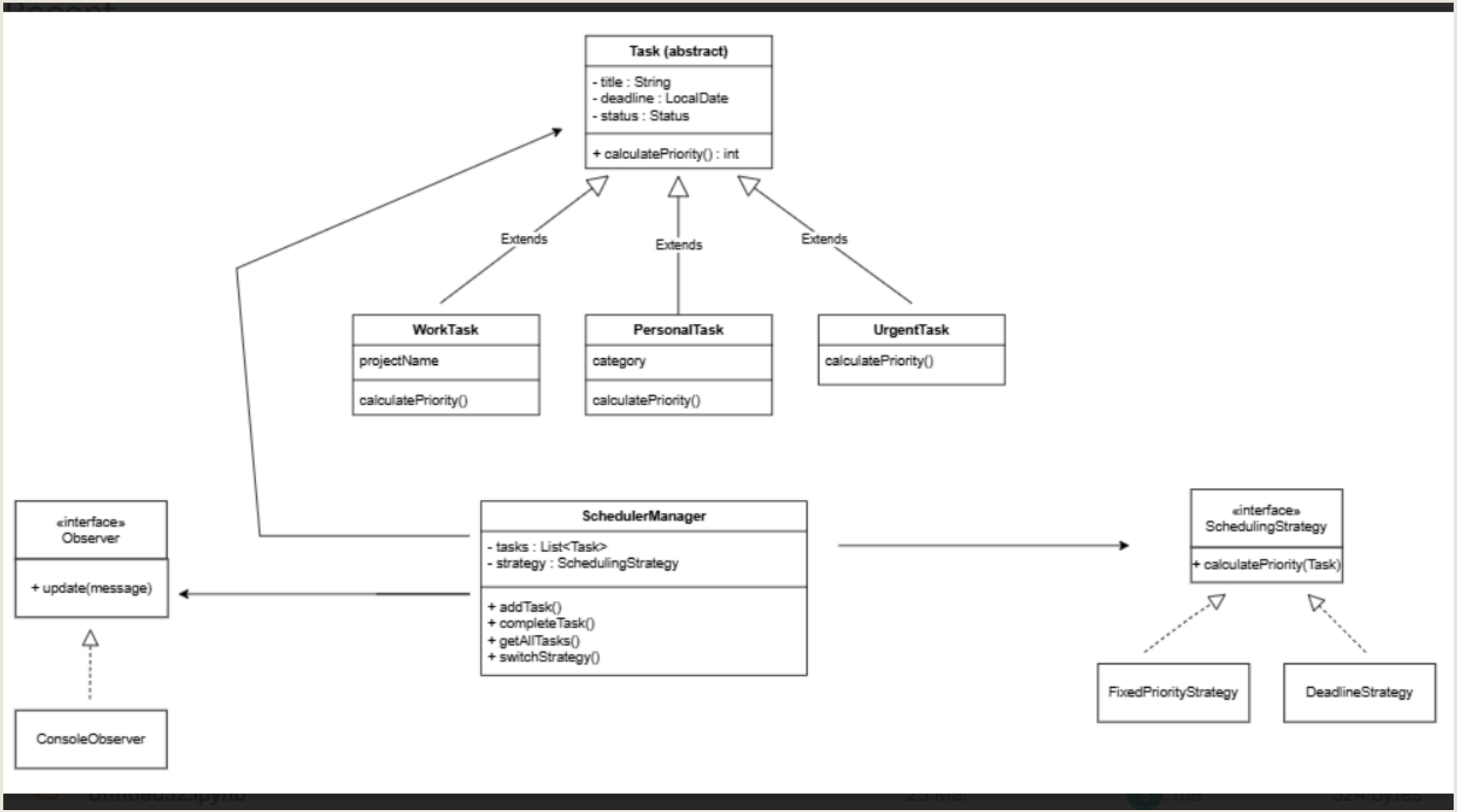
# Smart Task Scheduler with Dynamic Priority Management

**Problem Statement:** Static task managers fail to adapt to changing priorities and deadlines.

**Target Users:** Students, developers, researchers  and professionals managing multiple tasks.

# SYSTEM DESIGN




Architecture:
Modular Object-Oriented Design

Core Components:

- Task (Abstract Class)
- WorkTask, PersonalTask, UrgentTask
- SchedulerManag
- SchedulingStrat y (Interface)
- Observer Patter

Packages:

- model
- strategy
- manager
- observer

# OOP CONCEPTS USED

- Abstraction:
- Task abstract class defines common structure
- Inheritance:
- WorkTask, PersonalTask, UrgentTask extend Task
- Polymorphism:
- Scheduling strategies switched at runtime
- Encapsulation:
- Private attributes with controlled access (getters/setters)
- Exception Handling:
- Handles invalid input (e.g., incorrect date format)
- Collections:
- Uses ArrayList for dynamic task management
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Design Patterns:

- Strategy Pattern: Enables dynamic scheduling (Fixed / Deadline-based)
- Observer Pattern: Notifies system when tasks are added or completed

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# TECHNICAL HIGHLIGHT: DYNAMIC SCHEDULING SYSTEM

Core Idea: Strategy Pattern Implementation

- Decouples priority logic from task objects
- Supports runtime switching of scheduling algorithms

Implemented Strategies:

- Fixed Priority Strategy
- Deadline-Based Strategy

# System Execution

```
Main

--- Smart Task Scheduler ---
1. Add Task
2. View All Tasks
3. Complete Task
4. Switch Scheduling Strategy
5. Exit
Enter choice: 1
Task type (Work/Personal/Urgent): Urgent
Title: Submit Report
Task added: Submit Report

--- Smart Task Scheduler ---
1. Add Task
2. View All Tasks
3. Complete Task
4. Switch Scheduling Strategy
5. Exit
Enter choice: 2

All Tasks:
Pay Electricity Bill | Status: PENDING | Deadline: null | Priority: 10
Submit Report | Status: PENDING | Deadline: null | Priority: 10
Assignment | Status: PENDING | Deadline: null | Priority: 8
Buy Groceries | Status: PENDING | Deadline: 2026-04-05 | Priority: 5
Finish Report | Status: PENDING | Deadline: null | Priority: 8

--- Smart Task Scheduler ---
```

```
un   Main

1. Add Task
2. View All Tasks
3. Complete Task
4. Switch Scheduling Strategy
5. Exit
Enter choice: 3
Enter task title to complete: Assignment
Task completed: Assignment

--- Smart Task Scheduler ---
1. Add Task
2. View All Tasks
3. Complete Task
4. Switch Scheduling Strategy
5. Exit
Enter choice: 2

All Tasks:
Pay Electricity Bill | Status: PENDING | Deadline: null | Priority: 10
Submit Report | Status: PENDING | Deadline: null | Priority: 10
Finish Report | Status: PENDING | Deadline: null | Priority: 8
Buy Groceries | Status: PENDING | Deadline: 2026-04-05 | Priority: 5

--- Smart Task Scheduler ---
1. Add Task
2. View All Tasks
3. Complete Task
```

- Step 1: Add Tasks

Create Work, Personal, and Urgent tasks

- Step 2: View Tasks

Display tasks with priority and status

- Step 3: Switch Strategy

Change from Fixed → Deadline-based scheduling

- Step 4: Complete Task

Update task status to Completed
Observer displays notification

- Step 5: Error Handling

Handles invalid input (e.g., wrong date format)